

1 Introduction

TBD

2 Equations of Motion

Consider an infinite uncurved hyperplane $H \subseteq \mathbb{R}^3$ of slope $s \in \mathbb{R}$ and a rigid solid sphere of radius R and mass m in contact with the plane. Additionally, the sphere behaves as ideal magnetic dipole with dipole moment $\boldsymbol{\mu}_0$. The configuration manifold of the sphere is $Q := H \times \text{SO}(3) \subseteq \mathbb{R}^3 \times \text{SO}(3)$, and hence the velocity phase space is

$$TQ = TH \times T\text{SO}(3) \cong \mathbb{R}^2 \times \mathfrak{so}(3) \cong \mathbb{R}^2 \times \mathbb{R}^3 \subseteq \mathbb{R}^6.$$

Choose two global frames of $\mathbb{R}^3$, one constant frame $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ whose origin defines the position of the sphere at $t = 0$, and one moving frame $\{\hat{\mathbf{e}}_x(t), \hat{\mathbf{e}}_y(t), \hat{\mathbf{e}}_z(t)\}$ attached to the center of mass (COM) of the sphere. We will first derive the general equations of motions in $\mathbb{R}^3 \times \text{SO}(3)$ followed by transforming the equations to an H -adapted basis. All vectors measured with respect to the COM-frame will be denoted with a hat.